

Computer Vision SS26 - Exercise 2

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This exercise will be discussed the week of 11th of May.

1 Introduction

The following exercises cover theoretical aspects of image filtering, convolution, and linear shift-invariant operators.

2 Exercises

Exercise 1: Linearity and Shift Invariance

Let $f : \mathbb{Z}^2 \rightarrow \mathbb{R}$ be a discrete grayscale image and let $h : \mathbb{Z}^2 \rightarrow \mathbb{R}$ be a finite convolution kernel. The filtered image is defined as

$$g(y, x) = (f * h)(y, x) = \sum_{i, j} f(y - i, x - j)h(i, j).$$

Show that convolution with a fixed kernel h is a linear operator, i.e.

$$(f_1 + \lambda f_2) * h = f_1 * h + \lambda(f_2 * h),$$

for arbitrary images f_1, f_2 and scalar $\lambda \in \mathbb{R}$.

Furthermore, prove that convolution is shift invariant.

Exercise 2: Box Filter and Frequency Suppression

Consider the normalized 3×3 box filter

$$h = \frac{1}{9} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Let the image contain an impulse at the origin,

$$f(y, x) = \begin{cases} 1, & (y, x) = (0, 0), \\ 0, & \text{otherwise.} \end{cases}$$

Compute $(f * h)(y, x)$ explicitly.

Explain mathematically why this filter reduces sharp image structures. Relate your explanation to local averaging and the attenuation of high-frequency image components.

Exercise 3: Sharpening as Residual Filtering

Let $b = f * h$ be a smoothed version of an image f , where h is a normalized low-pass filter. A sharpened image can be defined as

$$g = f + \alpha(f - b),$$

where $\alpha > 0$.

Rewrite g as a single convolution operation

$$g = f * k$$

by deriving the equivalent kernel k in terms of h and the discrete delta kernel δ .

Discuss under which conditions this operation preserves constant image regions, and explain why it enhances edges.